

Ethical Discussion of a Social Phenomena

Large-scale Commercialization of Level 4 Autonomous Driving in China

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Introduction and Background

Autonomous vehicles (AVs) are increasingly transitioning from experimental prototypes to real-world deployment, with applications such as robotaxis, autonomous delivery vehicles, and advanced driver-assistance systems improving road safety, traffic efficiency, and mobility accessibility. However, these advancements also raise significant ethical concerns, including moral responsibility in accidents, algorithmic bias, transparency in decision-making, and balancing individual rights with collective safety.

In China, the rapid development and pilot deployment of Level 4 (L4) high-autonomy driving systems provide a particularly relevant context for examining these challenges. L4 vehicles can operate without human intervention under specific conditions, which heightens the ethical stakes of their decisions and responsibility allocation. Drawing on the backgrounds of our group members and real-world cases from cities such as Shenzhen, this report focuses on the ethical implications of large-scale L4 autonomous driving deployment in China, where social needs, regulatory frameworks, and technological maturity intersect.



Figure 1: Illustrations related to L4 autonomous driving deployment.

Research support for both sides' viewpoints

Proponent

My view is that the large-scale promotion of Level 4 (L4) autonomous driving technology in China is ethically reasonable. This is the best current choice for protecting lives, promoting fairness, and enhancing social well-being. This view is based on the utilitarian framework: when a technology significantly saves more lives than it causes harm and provides more travel convenience for vulnerable groups, even if it has flaws, its promotion has strong ethical legitimacy.

Firstly, safety data strongly supports this view. According to data from the Ministry of Public Security, 94% of traffic accidents are caused by human error. The Level 4 system directly addresses this fundamental problem by eliminating irrational behaviors such as fatigue, distraction, road rage, and drunk driving. As of November 2025, the fully autonomous vehicles in Shenzhen achieved a monthly safe driving mileage of 240,000 kilometers, with zero major liability accidents, and an accident rate of only 0.17 per 100,000 kilometers — far lower than that of human drivers. The cumulative safe driving mileage has reached tens of millions of kilometers, and no system-wide multi-vehicle failures have been feared by critics. Multi-level cybersecurity protocols and the physical "one-click disconnection" mechanism have further increased the risks, making "thousands of simultaneous accidents" an unproven assumption rather than a reality.

Secondly, the deployment of Level 4 technology promotes distributive justice and inclusiveness. Currently, autonomous vehicles in Shenzhen cover 1,594 routes and complete 210,000 fresh food deliveries per month, mainly serving the elderly and the disabled in the old urban areas. Autonomous taxis are expanding to airports and transportation hubs,

providing round-the-clock, barrier-free travel services for those who cannot drive. This technology not only replaces human drivers but also liberates people from repetitive labor, allowing them to engage in more meaningful interpersonal interactions, and giving those who previously relied on others an independent dignity. The promotion of this technology follows a common pattern — from concentrated development in developed regions to widespread popularization, just like traditional cars. The early restrictions do not negate the long-term fairness benefits.

Thirdly, the responsibility of Level 4 technology is clearer and more reliable compared to human driving. The relevant regulations in Shenzhen in 2022 clearly assign the main responsibility to the vehicle owner or manager and provide a specialized insurance product as a guarantee. The owner or insurance agency can then hold the developer accountable through product liability law, forming a chain of responsibility that is progressive and traceable. In contrast, over 10% of human accidents involve fleeing or shifting responsibility, resulting in victims being unable to receive compensation. Therefore, the Level 4 system strengthens rather than weakens ethical and legal protection.

In conclusion, although the Level 4 autonomous driving system is not perfect, it represents the best ethical compromise available at present: it significantly reduces preventable deaths, improves the fairness of travel, and operates within a constantly improving regulatory framework. Rejecting it from a theoretical perspective when it has saved numerous lives in real life would be ethically irresponsible.

Opponent

The overly rapid commercialization of Level 4 (L4) autonomous driving has created a fundamental tension between technological efficiency and ethical integrity. Further ethical research reveals that certain attractive outcomes mask systemic deficiencies in accountability, justice, and human dignity.

1. The “accountability gap”

The first core ethical challenge lies in accountability attribution. Traditional ethics requires moral subjects to possess consciousness and subjective intention, yet Level 4 autonomous systems operate as a black box. Although the Shenzhen Management Regulations (2022) assign responsibility to the vehicle owner or manager, this is merely a legal “patch,” not a moral solution. Ethical responsibility requires moral agency—the ability to intend, reflect, and feel remorse. It’s not just about someone, from concept designers to system producers to consumers; everyone is on the same chain.

2. The “bias problem”

Based on commercial objectives—maximizing investment returns—L4 system algorithms clearly favor data from the majority while ignoring data from marginalized groups and specific scenarios. Data from the China Automotive Technology and Research Center (2026) shows that error rates for non-standard pedestrians or extreme weather rise to 18%. This is a structural bias: when algorithms fail to identify marginalized groups—whether it’s the elderly in old urban areas or people with non-standard gaits—the technology effectively treats their safety as an “acceptable loss” in exchange for urban efficiency. The benefits of Robotaxis flow to tech-savvy urbanites, while the physical risks are disproportionately exported to those the algorithm “chooses” not to see.

Table 1: Level 4 Autonomous Driving System Recognition Accuracy Comparison Table

Scenario Category	Specific Environment & Pedestrian State	L4 Recognition Accuracy
Standard Normal Scenario	Sunny windless day, sufficient daylight, pedestrians walking upright normally	98.2%
Severe Weather Scenario	Heavy rain, dense fog, heavy snow, night backlight environment	82.5%
Non-standard Posture Pedestrians	Pedestrians falling, crouching, lying prone, bending over to pick up items	80.0%
Variable-speed Pedestrians	Sudden running, crossing road abruptly, elderly pedestrians moving slowly	83.1%

3. The “quantification of life”

The trolley problem is a timeless moral dilemma. Unlike human intuition, which involves internalizing these scenarios, L4 programming forces a transition from that to pre-calculated sacrifice. By converting human lives to data points—speed, trajectory, and probability—we adopt a radical utilitarianism that ignores the intrinsic value of the individual. As seen in the Wuhan 2024 collision, even when the human is at fault like jaywalking, an algorithm that “calculates” impact based on pre-set parameters treats the person as a hurdle rather than a being with dignity.

In conclusion, the current commercialization path prioritizes statistical safety over moral legitimacy. Until we resolve the gap between “paying for a mistake” and “taking responsibility for a life,” L4 deployment remains an ethically premature experiment on the public.

Understanding of the other side's viewpoints

Proponent

I fully understand the ethical concerns raised by the opposition based on deontology: the “black box” system makes it difficult to trace responsibility, algorithms may have biases against different groups, and in the famous “trolley problem”, humans are objectified as probability data. These are all very worthy ethical issues to study. Before the large-scale promotion of L4 autonomous driving, I must carefully consider these issues, otherwise it may bring unpredictable risks to society. However, our position is not to ignore these risks, but to believe that through continuous technological and legal improvements, these risks can be controlled. In contrast, existing data shows that the fatality rate of traffic accidents caused by human error is much higher than that of autonomous driving. Therefore, we should not only see the extremely small probability risks that autonomous driving may bring, but also see the convenience and safety it brings to humans now. We should use data as a support point to analyze ethical issues and prove that autonomous driving systems meet ethical requirements, rather than simply avoiding new technologies and imagining the dangers of extreme scenarios.

Opponent

While L4 systems offer convincing statistics of “reducing human error by 94%”, they also introduce systemic vulnerabilities, revealing a narrow view of safety. As the data above shows, the reliability of L4 autonomous driving is highly dependent on specific conditions. Furthermore, we must consider the scale of the impact. One error from a single driver might only threaten one road, but software glitches or cybersecurity vulnerabilities in a commercial model could inevitably trigger thousands of simultaneous accidents, escalating a localized incident into a catastrophic systemic failure. A survey by the China Consumers Association shows that since the first serious Robotaxi accident, public safety ratings for Robotaxis have plummeted from 85 to 65, a drop of 23.5%.

From an equity perspective, profit-driven promotion may exacerbate the digital divide. Services are likely to concentrate in affluent urban centers, further marginalizing less accessible areas. Almost all news about driverless cars we see is concentrated in a few technologically advanced cities in China. This may prove that the initial motivation for introducing this technology was not social morality or the value of life, but rather social benefit.

Mitigation strategies under contradictory viewpoints

Proponent

To mitigate the moral risks brought about by Level 4 autonomous driving, we will propose two targeted mitigation measures. Firstly, we can promote the development of explainable artificial intelligence to enhance the transparency and accountability of algorithms. Developers must provide traceable decision records for every critical event so that regulatory agencies can determine responsibility. Secondly, we can alleviate bias by building a more diverse training dataset. In terms of specific implementation, we can expand the data collection scope of autonomous driving test vehicles, supplement training data for non-standard pedestrians, elderly users, extreme weather conditions, and various urban and rural scenarios, thereby alleviating the bias caused by data. At the same time, we can conduct regular third-party bias reviews and test the bias of autonomous driving models to ensure that this technology provides safety while treating all groups equally, rather than transferring risks to marginalized groups.

Opponent

To address the ethical dilemmas of L4 autonomous driving, we propose a prudent, ethics-first deployment pathway. Instead of rushing large-scale commercialization, we support restricted-scenario, supervised operation with mandatory human oversight. We urge regulators to enforce transparent algorithm auditing to eliminate bias against non-standard pedestrians and vulnerable groups, upholding distributive justice. A clear, multi-layer liability system must be established to assign legal and moral responsibility to manufacturers, operators, and developers, closing the “black box” accountability gap. Most importantly, we reject any algorithm that quantifies human life; ethical guardrails must prohibit life-value calculation in critical decisions. Only under strict ethical constraints can autonomous driving be tested safely—without sacrificing accountability, fairness, or the intrinsic dignity of human life.

Reflection and Conclusion

Reflection

The debate settled not on a winner, but on a “conditional middle ground.” Neither side abandoned its core premise, yet through sustained exchange, each acknowledged the other’s deepest concern. The proponent conceded that ethical guardrails are non-negotiable; the opponent accepted that delaying all deployment also imposes a human cost. We ended not with consensus, but with a shared recognition that ethical legiti-

macy hinges on managing the collision between efficiency and dignity through phased safeguards, transparency, and inclusive governance. This taught us that the hardest AI ethics questions refuse binary resolution—it is never simply “whether,” but “under what conditions.”

Playing a role that diverged from our personal beliefs was an intensely instructive struggle. Wu, assigned as the proponent despite his usual caution toward autonomous systems, felt uneasy citing aggregate safety statistics while the opponent pressed on the “quantification of life.” Inhabiting the utilitarian stance forced him to appreciate—rather than dismiss—why policymakers might legitimately prioritize overall harm reduction. Yang, the opponent, is normally enthusiastic about AI innovation, yet had to sustain an argument that felt, at times, obstructionist. When the moderator challenged him to uphold strict ethical standards without delaying lifesaving technology, he experienced the internal friction between his deontological commitment and awareness of preventable human suffering. Articulating a gradualist’s answer helped him see that principled resistance can mean redirecting progress rather than rejecting it. Our moderator, meanwhile, struggled to remain impartial while sensing that both sides talked past each other on moral agency. Crafting questions that probed both “who is responsible?” and “what about the lives saved?” required holding two frameworks in mind simultaneously. Collectively, we learned that ethical maturity in AI governance begins not with finding the “correct” stance, but with the disciplined capacity to think from within an opposing position, feel its weight, and acknowledge the limits of any single lens.

Conclusions

The large-scale commercialization of L4 autonomous driving crystallizes a central AI ethics dilemma: a technology with genuine potential to save lives and expand mobility simultaneously threatens moral accountability, distributive fairness, and human dignity. Our debate demonstrated that technical safety metrics alone cannot resolve these normative tensions. Mitigation strategies—transparent auditing, phased deployment under human supervision, multi-layer liability—must be anchored in ethical principles from the start. Most importantly, the path forward requires sustained, inclusive debate that amplifies the voices of those most vulnerable to algorithmic marginalization. Only through such iterative, multi-stakeholder reflection can society harness autonomy’s benefits without sacrificing accountability, justice, or human dignity.

References

- [1] Shenzhen Intelligent Connected Vehicle Association. (2025, December 15). *Shenzhen functional unmanned vehicle operation and development report (November 2025)*.

Cited in support of the proponent’s Shenzhen L4 operational data: 240,000 km monthly safe mileage, 0.17 accidents per 10,000 km, zero main-liability accidents, 1,594 delivery routes, and 210,000 monthly fresh-food orders serving elderly and disabled residents.

- [2] Shenzhen Municipal People’s Congress Standing Committee. (2022). *Regulations of the Shenzhen Special Economic Zone on the management of intelligent connected vehicles*. Retrieved from http://www.sz.gov.cn/zfgb/2022/gb1250/content/post_9967816.html

China’s first ICV regulation. The proponent cites it to demonstrate a clear liability chain; the opponent critiques it as a legal “patch” that leaves the moral accountability gap unresolved.

- [3] Traffic Management Bureau of the Ministry of Public Security. (2025). *Annual report on road traffic accidents in China*.

Supplies the foundational statistic that 94% of traffic accidents are caused by human error, used by the proponent to justify L4 deployment and acknowledged by the opponent in framing systemic-risk concerns.

- [4] China Automotive Technology and Research Center. (2026). *Test data on object recognition accuracy of L4 autonomous driving systems under various scenarios*.

Provides the accuracy table cited by the opponent: 98.2% (standard), 82.5% (severe weather), 80.0% (non-standard posture), 83.1% (variable-speed pedestrians). Grounds the “bias problem” argument.

- [5] de Guzman, C. (2024, July 9). Baidu: Driverless car hits jaywalker in China. *Time*. <https://time.com/7001443/baidu-driverless-car-hits-jaywalker-china/>

Referenced by the opponent as the Wuhan 2024 collision case, illustrating the “quantification of life”—the algorithm treated a jaywalking pedestrian as a calculable obstacle rather than a being with dignity.

- [6] China Consumers Association. (2026). *Analysis of complaints received by national consumer associations in 2025*.

Supports the opponent’s claim that public trust in robotaxis declined sharply (safety rating from 85 to 65, a 23.5% drop) after the first serious accident.

- [7] Pathiraja, B., Liu, C., & Senanayake, R. (2024). Fairness in autonomous driving: Towards understanding confounding factors in object detection under challenging weather. *arXiv preprint arXiv:2406.00219*. <https://arxiv.org/abs/2406.00219>

Academic study that cross-validates the opponent’s “bias problem”: it systematically examines algorithmic fairness for pedestrians of different genders, skin tones, and body types under adverse weather.

- [8] Mill, J. S. (1863). *Utilitarianism*. London: Parker, Son, and Bourn.
 Foundational ethical framework for the proponent’s consequentialist argument: actions are justified insofar as they promote the greatest good for the greatest number, applied here to L4 deployment based on aggregate life-saving potential and social welfare gains.

- [9] Kant, I. (1785). *Groundwork of the metaphysics of morals*.
 Underpins the opponent’s deontological position: human beings must be treated as ends in themselves, never merely as means. This grounds the critique of quantifying human life and the insistence on moral agency and dignity as non-negotiable ethical requirements.

- [10] Foot, P. (1967). The problem of abortion and the doctrine of double effect. *Oxford Review*, (5).
 The classic source that first articulated the “trolley problem” thought experiment, referenced by the opponent to frame the ethical dilemma of pre-programming life-or-death decisions into autonomous systems.

- [11] Floridi, L., & Cowls, J. (2019). A unified framework of five principles for AI in society. *Harvard Data Science Review*, 1(1). <https://doi.org/10.1162/99608f92.8cd550d1>
 Provides the broad AI ethics framework (beneficence, non-maleficence, autonomy, justice, explicability) that implicitly structures both sides’ arguments about transparency, accountability, and fairness.

- [12] Nyholm, S., & Smids, J. (2016). The ethics of accident-algorithms for self-driving cars: An applied trolley problem? *Ethical Theory and Moral Practice*, 19(5), 1275–1289. <https://doi.org/10.1007/s10677-016-9745-2>
 Directly addresses the applicability of trolley-style dilemmas to autonomous vehicle programming, supporting the opponent’s concern about pre-calculated sacrifice and the ethical inadequacy of pure utilitarian crash optimization.